Heuristic Analysis Report for **Denton Public Library Catalog**

Executive Summary

This report contains an analysis of the Denton Public Library website considering certain parameters like Design, Content, Navigation, etc. and recommendations based on those analyses. The method of analysis in this case was a heuristic evaluation, which involved a website design expert analyzing the library website according to certain criteria. The criteria can be found in the Appendix.

There were many positives from the analysis, which included a strong navigation section and excellent search capabilities. The analysis also indicated certain flaws with the website. The lack of Contact Us information by way of a button was the biggest flaw of the website.
The website could be greatly improved by slight modifications which could make for much better user experience. Some of the recommendations are:

- Inclusion of Contact Us on the Home Page
- Alignment of graphics
- Increment in logo size
- Similar size and style of the Go button
- Descriptive labels for tabs

There are certain limitations to a heuristic analysis. This include the difficulty for a single person in identifying all problems with a website, as well as the subjectivity of a single person's opinion. All these can be counter indicated by performing other usability tests like a Card Sort or a Think Aloud protocol.

Methodology

Site Choice

The site I chose was the Denton Public Library catalog, located at https://library.cityofdenton.com/, since I thought there were obvious issues with the design of the website.

Objectives

The objectives of this Heuristic Analysis were manifold. They were as follows:

- Create a checklist to analyze the website.
- Analyze the website according to the checklist, which includes various metrics included in the next section.
- Find positive and negatives on the site using previous experience of heuristic evaluation.
- Suggest solutions to the negatives and provide positive feedback to the positives.

Metrics

There are a number of metrics used to analyze websites. I used these overarching metrics to analyze the Denton Library Catalog Website:

1. Home Page
2. Navigation
3. Design
4. Content
5. Search
6. Graphics and Images
7. Text
8. Consistency

I asked myself certain questions while analyzing the site, and I shall include the link to these in the appendix, since it is a very large file.

The four criteria for deciding these questions are as follows:

Major Problem. This problem would prevent nearly all users from completing their task, would cause users to have a misunderstanding of the purpose of a website or a task, and would cause users to abandon the site.

Moderate Problem. This problem may not be catastrophic but would cause an issue for many users, or would cause a catastrophic problem for a handful of users.

Cosmetic Problem. This problem likely will not prevent users from doing a task, but may delay them or cause them to have an unhappy user experience.

No Problem. The issue is not a problem in the website.

Evaluation Tasks

I have listed the main evaluation parameters from every criterion below, so that the findings are easily understood. I went through the site based on the list in the Parameters section the Appendix and listed the positives and the shortcomings in the findings.

Home Page

1. Download Time for page
2. Company/Brand About Us
3. Contact Us link
4. Creation of Positive Impression
5. Home Page looks like Home Page
6. Access to Home Page from other Pages

Navigation

1. Navigation capability on each page
2. Clickable list of contents on long pages
3. Navigation bar items < 9

Design

1. Liquid Layout
2. Page Length
3. Logo Placement/Size
4. Background Color
5. White Space
6. Content Alignment
7. Line Length

Content

1. Useful Content
2. No Unsolicited Windows/Graphics
3. Meaningful Hyperlinks
4. Descriptive Labels
5. Use of familiar words

Search

1. Search function
2. Placement of Search

Graphics and Images

1. Meaningful graphics
2. Images slowing page download
3. Good contrast between background and text

Text

1. Use of good-size font

Consistency

1. Typographical consistency
2. Graphical consistency
3. Consistency of color
4. Consistency of buttons and icons

Each of these criteria shall have the colored icon next to it to show if it was a problem or not. Any positives and negatives shall be justified with comments relevant to the site.

Shortcomings

The possible shortcomings of this analysis are as follows:

- I might be biased towards a certain layout, and all users might not agree with that. However, I did my best to accommodate all users.
- Some users might be used to the current layout, and that might affect the library wanting to change the website.
- Library catalog websites are unlike most other websites, in that basic usability is favored over design choices.

Other usability tests like think-aloud protocols could be conducted to justify the veracity of this study.

Main Findings

The crucial findings are included here. The others are reported in the Appendix, located at the end of this document.

Home Page

Positives

Download Time for page
The page downloads very quickly, in less than a couple of seconds. That is always a good sign.

Access to Home Page from other Pages
There is good access to the Home Page from other pages, because clicking on the Denton Public Library logo takes you to the Home Page of the website.

Negatives

Contact Us link
I think that this is the biggest problem with the Library website. There is no indication of how to contact the library, even if you look at all the tabs and options. You can only contact the library by clicking on Library Website, which takes you to the departments of Denton Government and then you navigate to details about the library. That is inconvenient for anyone who wants to find the contact details for the library quickly.

Navigation

Positives

Navigation capability on each page
There are good navigational capabilities on each page. For long pages, there is the Back to Top button, which is always useful. The navigational bar stays in its place on every page.

Navigation bar items < 9
There are 8 buttons on the main navigation bar, with drop down lists populating each button.

Negatives

Navigationally, there were no glaring, obvious errors. Maybe the tabs could be renamed so as not to confuse patrons, but since scrolling is fast, this is not an issue.

Design

Positives

Liquid Layout
There is a liquid layout to the website, which is useful in this day of mobile computing. Websites shouldn't lose any functionality because of the inability of sites to adjust to screen resolutions and sizes.

Negatives

Content Alignment
Content alignment could be better on the home page, which might improve usability.

Logo Placement/Size
The logo on the Home Page is much smaller than some of the graphics, so that should be remedied. The logo is well designed, so making it larger might be a good thing.

Content

Positives

No Unsolicited Windows/Graphics
The website doesn't force any popups or graphics on you.

Negatives

Descriptive Labels
The labels need more description, since they don't define anything clearly enough for the user.

Search
Positives

Search function
The search function is good, but it doesn't obviously say Search anywhere. The Search functionality should be explicitly mentioned.

Placement of Search
It isn't placed on the obvious top right like most other search bars, but like Google, it is placed in the center, which is a huge positive.

Graphics and Images
The page scores positives on all of the factors mentioned above. However, the library might want images to supplement all that text.

Text
The text size might be an issue at 11 point, but it is still readable. However, if the library feels that its audience is going to be the older crowd, it should increase font size.

Consistency
Positives

Typographical consistency
The typography is remarkably consistent throughout the website.

Consistency of color
The background and foreground color are consistent throughout the website.

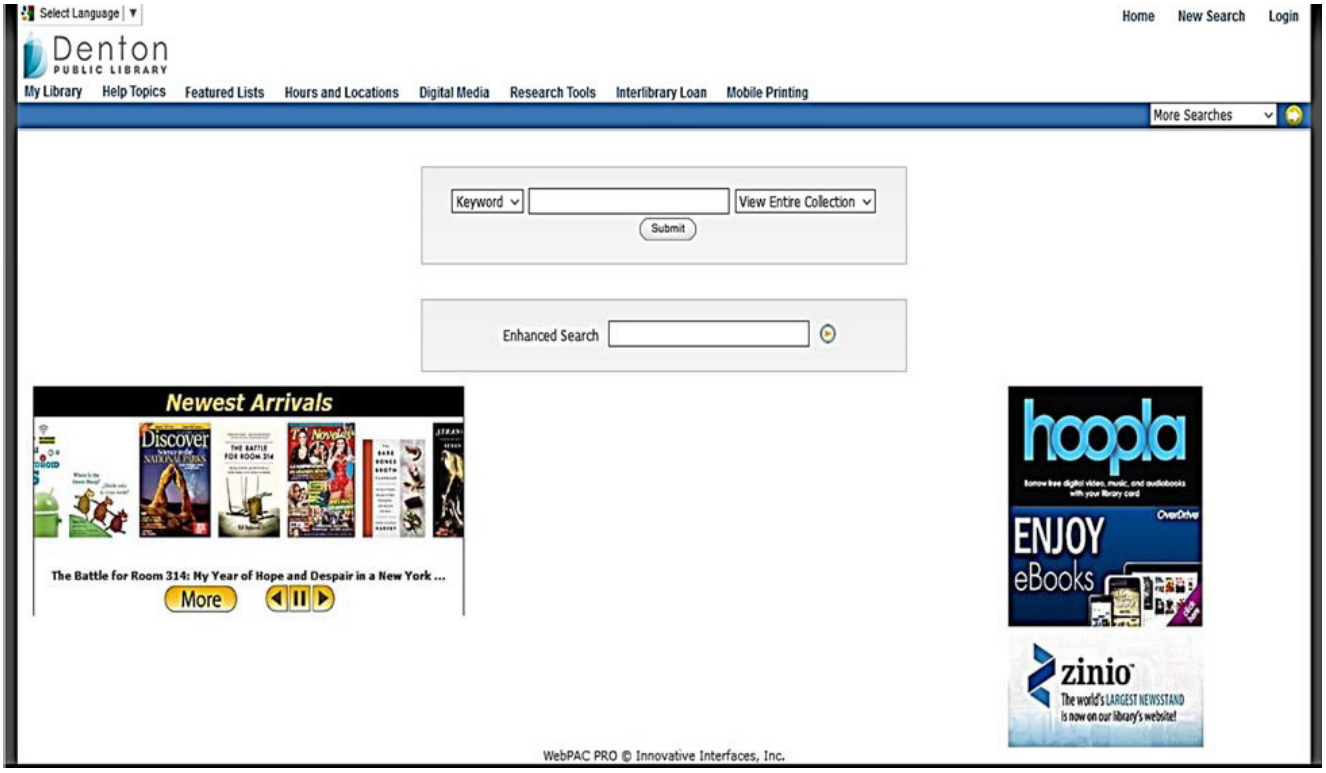
Negatives

- **Consistency of buttons and icons**
  The Go buttons are different on the Home Page itself. This is bound to confuse the patron and will make for bad usability.

Recommendations

The main recommendations would be as follows:

- Include Contact Us information on the Home Page, at its general location in the footer. Contact Us helps in patrons feeling that the library is welcoming them to ask for info.
- The alignment and resizing of the graphics is also essential, since the current page looks lopsided. Maybe the graphics could be adjusted side by side so as to occupy the large amounts of white space that exist in the middle.
- The logo should be made much bigger, since that is an important part of the page. It should at least bigger than the Digital Media logos on the bottom right. The logo is well designed and could benefit from a resizing.



- The Go buttons on the Home Page should be made similar to each other, since they don't look alike at all right now. The way they exist currently might create a confusion among patrons who might think that those aren't Go buttons.

- The labels in the navigation bar and each drop down list should be more descriptive. For example, Searching, which is a member of the list of Help Topics should be expanded to say Search Tips, which might be a useful of telling people how to search.

The other recommendations, made by taking a look at some of the more cosmetic findings in the Appendix, are as follows:

- There needs to be an About page, but since it is a library website, that is easy for users to understand.
- Links which open as PDFs or which link to another site should clearly be tagged.

Conclusion

On the whole, the website is user friendly, but it has its fair share of problems. The recommendations can help overcome most of the glaring issues. There are a lot of positives to the website, as listed in the Findings. These should be kept as is, so that the library catalog is improved for future users.

It is possible that certain biases crept into this analysis, but the subjective nature of this study is one of its biggest benefits. Other studies should be carried out to compare the results of this study versus a Think Aloud protocol or a Card Sort.

The current website is in decent shape, but the few tweaks that need to be made are crucial, especially the Contact Us information.

Appendix

Parameters

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Additional Findings

Home Page

Company/Brand About Us
Since it is a Denton Public Library Catalog website and that is mentioned everywhere, it probably doesn't need a description of itself. But they might want to enumerate their services somewhere.

Creation of Positive Impression
The website, as a whole, does not create a very positive impression, but it does what a library website is supposed to do.

Home Page looks like Home Page
It is a barebones Home Page, but it does the job of a Library home page, in that it gives patrons a chance of searching for books and provides necessary tabs to scroll to important aspects of the library.

Navigation

Background Color
The background color is white, and this is desirable because it provides good contrast against black text.

Line Length
The line length is perfect, considering each line doesn't go above 50 words.

Design

Page Length
Some pages are long, but the clickable list of contents and the Back to Top mentioned in the previous unit always help, so it is a cosmetic change that might be implemented in the future.

White Space
Good use of white space in most places, but sometimes, it feels like there is too much white space on some pages. That might be down to the lack of graphics and images.

Content

Useful Content
Most of the content is useful. More images could be added to break the monotony in text.

Meaningful Hyperlinks
Each hyperlink is useful, and there's nothing that is going to blindside you. However, some of the tabs take you to PDFs or the Denton Government website, which should be avoided.

Use of familiar words
Some words like Interlibrary Loan and Friends of the Library might be unfamiliar to people not regularly in contact with libraries, but this isn't a glaring problem.

Consistency

Graphical consistency
The library might want to add more images, and make the existing graphics equal in size.